

Credit Default Swaps with Risky Counterparty. Model With Payoff Acceleration.

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1 Introduction

A credit default swap (CDS) is a default protection instrument written on a corporate bond. The buyer of the protection pays a specified rate on a specified notional principal until either the bond's issuer defaults or the contract expires. If the issuer defaults before the contract expiration, the buyer is entitled either to exchange the bond for its face value N or to receive a cash equivalent payment given by

$$(1 - R_B)N, \tag{1}$$

where R_B is the issuer's recovery rate, unless the protection seller defaults before the issuer, in which case the deal is marked to market and if the protection buyer owes to the protection seller, the payment is made in full, otherwise, the payment is multiplied by the protection seller's recovery rate R_C . Thus, there are two types of default risk associated with this type of contracts: risk of default of the issuer and risk of default of the protection seller.

In general, a CDS contract is characterized by: the maturity of the deal (*e.g.* 5 years), the coupon frequency (*e.g.* quarterly), the notional principal (*e.g.* \$10M), and the spread or coupon payment (*e.g.* 100bps). In addition, one must supply information about: the actual coupon schedule (*e.g.* first/last coupon dates, the date generation method), the day count convention (*e.g.* ACT/360), the business day calendar, and the payment dates offsets. We assume that each recovery rate is a known quantity supplied by the user (*e.g.* 0.4).¹

¹In general the recovery rates are known only when the default event is officially recorded, and its value may depend on many economic factors. To express this market reality, one can model the recovery rates as random variables. However, in this paper we follow the common practice of assuming deterministic recovery rates.

2 Pricing credit default swaps

To price a CDS we need to price the following three pieces: premium leg, default leg, and coupon payment accrued at default. Let us introduce the following notation:

- T - the deal's maturity,
- N_p - the number of coupon payments,
- c - the coupon rate value,
- τ_B - the time to default of the bond's issuer with $S_B(t)$, $F_B(t)$, $f_B(t)$ denoting its survival, cumulative distribution, and probability density functions respectively,
- τ_C - the time to default of the protection seller with $S_C(t)$, $F_C(t)$, $f_C(t)$ denoting its survival, cumulative distribution, and probability density functions respectively,
- T_j - the j -th coupon date,
- Δ_j - the year fraction between T_{j-1} and T_j ,
- Δ_t - the year fraction between t and the last coupon date before t ,
- Δ - the time (in years) it takes the protection seller to make the default payment,
- $D(t)$ - the discount factor from t to the evaluation time 0.

There are three possible situations. First, the protection seller can default at least Δ years after the issuer's default. In this case, we just have a regular CDS situation where the coupon payments are made until the issuer defaults, after which the accrued payment is made and the default payment of $N(1 - R_B)$ is received. If the protection seller defaults after the issuer but before Δ years period passes, the protection seller will only pay R_C fraction of the default payment. Finally, if the protection seller defaults before the issuer, the CDS deal is marked to market and if the protection buyer owes to the protection seller, the payment is made in full, otherwise the protection seller

pays only R_C fraction of the value to the protection buyer. Making this discussion formal, we can write

$$\begin{aligned}
Value &= \sum_{j=1}^{N_p} \int_{T_{j-1}}^{T_j} \{ c \sum_{k=1}^{j-1} D(T_k) P(\tau_B > T_k \mid \tau_C = t) \\
&+ c \int_0^t D(u) \Delta_u f_{\tau_B \mid \tau_C}(u, t) du \\
&+ \min(FwdCDS(t, j), R_C FwdCDS(t, j)) \\
&- N(1 - R_B) \int_0^{t-\Delta} D(u) f_{\tau_B \mid \tau_C}(u, t) du \\
&- N(1 - R_B) R_C \int_{t-\Delta}^t D(u) f_{\tau_B \mid \tau_C}(u, t) du \} f_{\tau_C}(t) dt \\
&+ CDS(\tau_C > T_{N_p}),
\end{aligned} \tag{2}$$

where

$$\begin{aligned}
FwdCDS(t, j) &= c \sum_{k=j}^{N_p} D(T_k) P(\tau_B > T_k \mid \tau_C = t) \\
&+ c \int_t^T D(u) \Delta_u f_{\tau_B \mid \tau_C}(u, t) du \\
&- N(1 - R_B) \int_t^T D(u) f_{\tau_B \mid \tau_C}(u, t) du.
\end{aligned} \tag{3}$$

Actually, what we should really do here is define

$$\begin{aligned}
FwdCDS(t, j) &= c \sum_{k=j}^{N_p} D(t, T_k) P(\tau_B > T_k \mid \tau_C = t) \\
&+ c \int_t^T D(t, u) \Delta_u f_{\tau_B \mid \tau_C}(u, t) du \\
&- N(1 - R_B) \int_t^T D(t, u) f_{\tau_B \mid \tau_C}(u, t) du,
\end{aligned} \tag{4}$$

where $D(T_1, T_2)$ denotes discount factor from T_2 to T_1 , and then write

$$\begin{aligned}
Value &= \sum_{j=1}^{N_p} \int_{T_{j-1}}^{T_j} \{ c \sum_{k=1}^{j-1} D(T_k) P(\tau_B > T_k \mid \tau_C = t) \\
&+ c \int_0^t D(u) \Delta_u f_{\tau_B \mid \tau_C}(u, t) du \\
&+ \min(FwdCDS(t, j), R_C FwdCDS(t, j)) \\
&- N(1 - R_B) \int_0^{t-\Delta} D(u) f_{\tau_B \mid \tau_C}(u, t) du \\
&- N(1 - R_B) R_C \int_{t-\Delta}^t D(u) f_{\tau_B \mid \tau_C}(u, t) du \} f_{\tau_C}(t) dt \\
&+ CDS(\tau_C > T_{N_p}),
\end{aligned} \tag{5}$$

but since $D(t)$ is always positive, and we assume no evolution in either interest rate or credit curves, we can simplify our formulas like this.

$$\begin{aligned}
CDS(\tau_C > T_{N_p}) &= c \sum_{k=1}^{N_p} D(T_k) P(\tau_B > T_k, \tau_C > T_{N_p}) \\
&+ c \int_0^{T_{N_p}} D(u) \Delta_u f_{\tau_B, \tau_C > T_{N_p}}(u) du \\
&- N(1 - R_B) \int_0^{T_{N_p}} D(u) f_{\tau_B, \tau_C > T_{N_p}}(u) du.
\end{aligned} \tag{6}$$

We assume that τ_B and τ_C are jointly distributed through two dimensional Gaussian copula, i.e.

$$\begin{aligned}
C_\rho(u_1, u_2) &= N_\rho(N^{-1}(u_1), N^{-1}(u_2)), \\
F_{BC}(t_1, t_2) &= C_\rho(F_B(t_1), F_C(t_2)) \\
&= N_\rho(N^{-1}(F_B(t_1)), N^{-1}(F_C(t_2))),
\end{aligned} \tag{7}$$

here N_ρ stands for a standard bivariate gaussian cdf with correlation ρ . Now,

we need to approximate integrals of the form

$$\begin{aligned}
I_1 &= \int_{T_{j-1}}^{T_j} D(u) f_{\tau_B|\tau_C}(u, t) du, \\
I_2 &= \int_{T_{j-1}}^{T_j} D(u) \Delta_u f_{\tau_B|\tau_C}(u, t) du,
\end{aligned} \tag{8}$$

and use numerical quadrature for the outside integration over τ_C .

3 Interpolation

We assume flat hazard rate for $\tau_B \mid \tau_C$ and flat forward rate on each interval $[T_{j-1}, T_j]$, i.e.

$$\begin{aligned}
D(t) &= D(T_{j-1}) e^{-f(t-T_{j-1})}, \\
S_{\tau_B|\tau_C}(t) &= S_{\tau_B|\tau_C}(T_{j-1}) e^{-h(t-T_{j-1})}, \\
f &= \frac{1}{\Delta_j} \ln \left(\frac{D(T_{j-1})}{D(T_j)} \right), \\
h &= \frac{1}{\Delta_j} \ln \left(\frac{S_{\tau_B|\tau_C}(T_{j-1})}{S_{\tau_B|\tau_C}(T_j)} \right),
\end{aligned} \tag{9}$$

Now, we can approximate the integrals we need to calculate:

$$\begin{aligned}
I_1 &= \int_{T_{j-1}}^{T_j} D(u) f_{\tau_B|\tau_C}(u, t) du \approx D(T_{j-1}) S_{\tau_B|\tau_C}(T_{j-1}, t) h \int_0^{\Delta_j} e^{-(f+h)u} du \\
&= D(T_{j-1}) S_{\tau_B|\tau_C}(T_{j-1}, t) \frac{h(1 - e^{-(f+h)\Delta_j})}{f+h} \\
&= \frac{h}{f+h} (D(T_{j-1}) S_{\tau_B|\tau_C}(T_{j-1}, t) - D(T_j) S_{\tau_B|\tau_C}(T_j, t)), \\
I_2 &= \int_{T_{j-1}}^{T_j} D(u) \Delta_u f_{\tau_B|\tau_C}(u, t) du \approx D(T_{j-1}) S_{\tau_B|\tau_C}(T_{j-1}, t) h \int_0^{\Delta_j} u e^{-(f+h)u} du \\
&= \frac{h}{f+h} D(T_{j-1}) S_{\tau_B|\tau_C}(T_{j-1}, t) \left(\frac{1 - e^{-(f+h)\Delta_j}}{f+h} - \Delta_j e^{-(f+h)\Delta_j} \right) \\
&= \frac{h}{(f+h)^2} (D(T_{j-1}) S_{\tau_B|\tau_C}(T_{j-1}, t) - D(T_j) S_{\tau_B|\tau_C}(T_j, t)) - \frac{h\Delta_j}{f+h} D(T_j) S_{\tau_B|\tau_C}(T_j, t).
\end{aligned} \tag{10}$$

$$\begin{aligned}
S_{\tau_B|\tau_C}(T, t) &= \frac{S_{\tau_B, \tau_C}(T, t)}{f_{\tau_C}(t)} = \frac{\frac{\partial}{\partial t} P(\tau_B > T, \tau_C < t)}{f_{\tau_C}(t)} = \frac{\frac{\partial}{\partial t} (P(\tau_C < t) - P(\tau_B < T, \tau_C < t))}{f_{\tau_C}(t)} \\
&= \frac{f_{\tau_C}(t) - \frac{\partial}{\partial t} F_{BC}(T, t)}{f_{\tau_C}(t)} = \frac{f_{\tau_C}(t) - \frac{\partial}{\partial t} N_\rho(N^{-1}(F_B(T)), N^{-1}(F_C(t)))}{f_{\tau_C}(t)} \\
&= 1 - \frac{\left(\frac{\partial}{\partial y} N_\rho \right) (N^{-1}(F_B(T)), N^{-1}(F_C(t)))}{\phi(N^{-1}(F_C(t)))} \\
&= 1 - \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}[N^{-1}(F_C(t))]^2} N \left(\frac{N^{-1}(F_B(T)) - \rho N^{-1}(F_C(t))}{\sqrt{1-\rho^2}} \right)}{\phi(N^{-1}(F_C(t)))} \\
&= N \left(\frac{\rho N^{-1}(F_C(t)) - N^{-1}(F_B(T))}{\sqrt{1-\rho^2}} \right),
\end{aligned} \tag{11}$$

$$\begin{aligned}
S_{\tau_B, \tau_C > T}(t) &= P(\tau_B > t, \tau_C > T) = P(\tau_B > t) + P(\tau_C > T) - 1 \\
&+ P(\tau_B < t, \tau_C < T) \\
&= S_B(t) + S_C(T) + N_\rho(N^{-1}(F_B(t)), N^{-1}(F_C(T))) - 1.
\end{aligned} \tag{12}$$

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